

A Modular Framework for Enhancing Satellite-Derived Daily Precipitation: Adjusting Values, Aligning Distributions, and Preserving Extremes

Seminar Draft

Benny Istanto^{1,*}

Rizaldi Boer^{2,3}

I Putu Santikayasa^{2,4}

¹ Applied Climatology Study Program, Department of Geophysics and Meteorology,
Bogor Agricultural University, Bogor 16680, Indonesia

² Department of Geophysics and Meteorology, Bogor Agricultural University, Bogor 16680, Indonesia

³ International Research Institute for Environment and Climate Change,
Bogor Agricultural University, Bogor 16680, Indonesia

⁴ Center for Climate Risk and Opportunity Management in South East Asia and Pacific,
Bogor Agricultural University, Bogor 16680, Indonesia

* Correspondence: bennyistanto@gmail.com

Abstract

Accurate daily precipitation estimates are essential for hydrological modeling, climate risk assessment, and disaster preparedness. Satellite-based products such as IMERG provide global coverage but exhibit systematic biases in daily accumulations, particularly in representing extreme precipitation events. This study presents a hybrid bias correction framework (LSEQM+DL) that sequentially integrates Linear Scaling (LS), Empirical Quantile Mapping (EQM) with Generalized Pareto Distribution (GPD) tail adjustment, and Convolutional Neural Network (CNN) refinement. A station density confidence mask modulates CNN influence based on gauge network density. Applied to daily IMERG Late Run over Indonesia (2001–2025) and evaluated against 171 independent BMKG station observations, the framework brings the standard deviation ratio from 0.71 (LS) to 1.00 (LSEQM+DL), the relative bias from -11.4% to -0.6% , and the 99th-percentile ratio from 0.71 to 1.01. The wet-day frequency ratio improves from 1.21 to 0.95, reducing a 21% overestimation of wet-day occurrence to within 5% of observed. Event detection (POD) decreases from 0.78 to 0.65, reflecting the trade-off inherent to distributional correction. WMO multi-threshold verification confirms maintained skill at thresholds up to 100 mm/day. The framework relies on globally available datasets and is applicable to other tropical regions with sparse gauge networks.

Keywords: Bias Correction; Satellite Precipitation; IMERG; Extreme Events; Deep Learning; Quantile Mapping; Generalized Pareto Distribution; Quality Assessment; Indonesia

1 Introduction

Accurate daily precipitation estimates underpin flood forecasting, agricultural planning, hydrological modeling, and climate risk assessment. Satellite-based products such as the Integrated Multi-satellite Retrievals for Global Precipitation Measurement (IMERG) provide near-global coverage [1, 2] but exhibit systematic biases relative to surface observations: overestimation of light-rain days, underestimation of heavy events, and distributional mismatch that degrades the representation of extremes [3]. These limitations are particularly consequential in tropical regions such as Indonesia, where convective rainfall dominates and ground gauge networks are sparse.

Several bias correction approaches address different aspects of this problem. Linear Scaling (LS) removes the mean bias but leaves the distribution shape unchanged [4]. Empirical Quantile Mapping (EQM) aligns the full distribution but can underrepresent the upper tail [5, 6]. Generalized Pareto Distribution (GPD) fitting targets extreme values explicitly [7] but is typically applied in isolation. Convolutional Neural Networks (CNNs) capture spatially complex residual patterns [9, 10] but risk overfitting in data-sparse regions. No prior study has integrated all four components into a sequential, physically interpretable framework for daily satellite precipitation correction.

This paper presents LSEQM+DL, a hybrid framework that addresses mean bias, distributional alignment, and extreme event preservation in sequence. Applied to daily IMERG Late Run over Indonesia (2001–2025) and validated against CPC-UNI gridded observations [13] and 171 independent BMKG station records, the framework yields substantial, independently verified improvements. The three design objectives provide the organizing structure for both the methods and the evaluation: *adjusting values*, *aligning distributions*, and *preserving extremes*.

2 Study Region, Data, and Methods

2.1 Study Region

Indonesia (95–141°E, 11°S–6°N) is the world’s largest archipelago, presenting a complex precipitation environment: a trimodal rainfall regime driven by the monsoon and Maritime Continent convective cycle [19, 20], frequent extreme events, and a gauge network heavily concentrated in Java and southern Sumatra. The 180 CPC-UNI contributing stations provide adequate constraint in western Indonesia but sparse coverage over Papua, Maluku, and eastern Nusa Tenggara. This spatial heterogeneity provides both a demanding test environment and a practical motivation for the station density confidence mask (Section 2.3.4).

2.2 Datasets

Three precipitation datasets are used. IMERG Late Run (IMERG-L; 0.1°, ~14-hour latency) [2, 21] is the product to be corrected; because it relies purely on satellite observations without gauge assimilation, it provides an uncontaminated baseline. IMERG Final Run (IMERG-F;

0.1°) [2], which incorporates gauge data during post-processing, is included as an independent benchmark. CPC Unified Gauge-Based Analysis of Daily Precipitation (CPC-UNI; 0.5°) [13, 18] is the correction reference. Independent validation uses 171 BMKG stations with at least 30 valid daily records per dekadal period (Figure 3).

Preprocessing harmonizes all datasets through nearest-neighbor regridding, temporal alignment, land-sea masking, and latitude reindexing. Following the Bias Correction Spatial Disaggregation (BCSD) principle [23], distribution parameters are estimated at the native 0.5° CPC-UNI resolution and interpolated to 0.1° before quantile mapping is applied. A wet-day threshold of 1.0 mm/day is applied throughout [24].

2.3 Hybrid Bias Correction Framework

The framework corrects daily satellite precipitation through four sequential stages, each targeting a distinct source of systematic bias (Figure 1). Processing is performed independently for each of 36 dekadal windows (three 10-day periods per month), with statistics pooled across all years (approximately 230 samples per dekad) for robust distribution fitting.

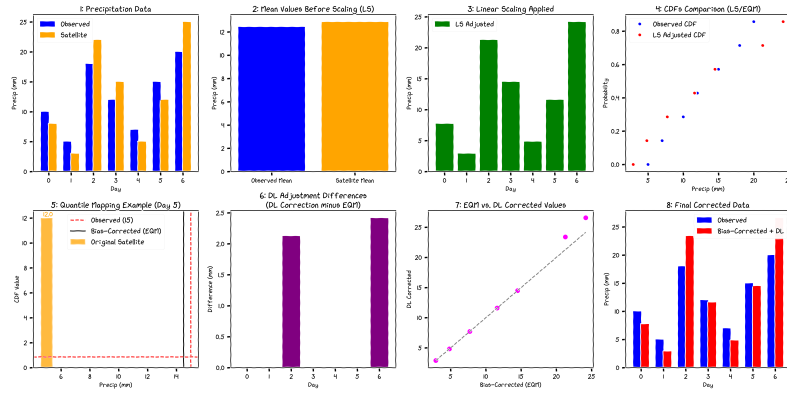


Figure 1: Conceptual workflow of the LSEQM+DL hybrid bias correction framework.

2.3.1 Stage 1: Linear Scaling

Linear Scaling removes the mean bias by multiplying each satellite field by a dekad-specific factor:

$$C_{LS} = \frac{\bar{P}_{ref}}{\bar{P}_{sat}}, \quad P_{LS} = P_{sat} \cdot C_{LS} \quad (1)$$

where $\bar{P}_{ref}$ and $\bar{P}_{sat}$ are the dekadal mean daily precipitation of CPC-UNI and IMERG-L, respectively. The scaling factor is computed at the native 0.5° CPC-UNI resolution and bilinearly interpolated to 0.1°, producing a spatially smooth correction field [23].

2.3.2 Stage 2: Empirical Quantile Mapping (EQM)

Empirical Quantile Mapping aligns the full cumulative distribution function (CDF) of the LS-corrected field to that of the reference. Gamma distributions are fitted to the wet-day sample using maximum likelihood estimation [15], and the quantile correction is:

$$P_{EQM} = F_{ref}^{-1}(F_{sat}(P_{LS})) \quad (2)$$

where F_{sat} and F_{ref} are the fitted satellite and reference gamma CDFs, respectively. Dry-day handling follows Cannon et al. [11], preserving the reference dry-day frequency by construction and preventing satellite drizzle from mapping to spurious low-intensity wet-day values.

2.3.3 Stage 3: GPD Tail Adjustment

Gamma-based EQM can underrepresent the upper tail of the precipitation distribution. Exceedances above the 80th percentile of the wet-day sample are instead modeled with the Generalized Pareto Distribution [7]. For an LS-corrected value P_{LS} exceeding the satellite threshold u_{sat} , the conditional exceedance probability and GPD-corrected value are:

$$p_c = \frac{F_{sat}(P_{LS}) - F_{sat}(u_{sat})}{1 - F_{sat}(u_{sat})}, \quad P_{LSEQM} = u_{ref} + G_{ref}^{-1}(p_c) \quad (3)$$

where G_{ref}^{-1} is the inverse GPD fitted to the reference exceedances and u_{ref} is the reference 80th percentile. GPD parameters are estimated at 0.5° and interpolated to 0.1° . Corrected values are capped at the interpolated 99.9th percentile of the reference distribution to prevent physically unrealistic extrapolation.

2.3.4 Stage 4: CNN Refinement and Spatial Modulation

Residual biases in extreme precipitation pixels, arising from complex spatial patterns that statistical methods cannot capture, are addressed by a CNN trained separately for each dekadal period with LSEQM-corrected fields as input and CPC-UNI as the training target. CNN corrections are applied selectively to extreme pixels through alpha-blending:

$$P_{final} = \alpha_{eff} \cdot P_{LSEQM} + (1 - \alpha_{eff}) \cdot P_{DL} \quad (4)$$

applied only to pixels where P_{LSEQM} exceeds the 80th percentile of the dekadal distribution; non-extreme pixels retain their LSEQM values. The effective blending weight is modulated by a station density confidence mask $C(x, y) \in [0, 1]$:

$$\alpha_{eff}(x, y) = 1.0 - C(x, y) \cdot (1.0 - \alpha) \quad (5)$$

where $\alpha = 0.70$ is the base weight. The confidence mask is derived from gauge counts per 0.5° cell, Gaussian-smoothed to prevent sharp boundaries, and normalized by a saturation count of 3 stations. Station-dense areas receive the full CNN contribution ($\alpha_{eff} = 0.70$); station-sparse areas revert toward pure LSEQM ($\alpha_{eff} \rightarrow 1.0$) [27].

2.4 Evaluation Approach

Performance is assessed through 31 pixel-level metrics organized by three evaluation pillars aligned with the design objectives: value adjustment (relative bias, standard deviation ratio), distribution alignment (Kolmogorov-Smirnov test, wet-day frequency ratio, wet-day intensity ratio, six percentile ratios), and extreme preservation (95th- and 99th-percentile ratios, WMO multi-threshold verification at seven thresholds from 1 to 150 mm/day). A Composite Quality Index (CQI) integrates these metrics into a single score on $[0, 1]$.

Evaluation employs two reference datasets: CPC-UNI (the correction target, for domain-wide assessment) and 171 independent BMKG station observations not used in the correction procedure (for ground truth validation). This two-reference design distinguishes genuine improvements from overfitting to the training reference.

3 Results

3.1 Domain-wide Performance

Table 1 summarizes the domain-wide spatial median across all land pixels. Linear Scaling reduces the relative bias against CPC-UNI to zero by construction, but the standard deviation ratio of 0.971 confirms that uniform multiplicative rescaling leaves the distribution shape essentially unchanged. The LSEQM stage reshapes the distribution: the wet-day-mean ratio increases from 0.961 to 1.157, the Q_{99} ratio from 0.967 to 1.212, and the wet-day frequency ratio decreases from 1.040 toward 0.947. The apparent overshoot of upper percentile ratios above unity against CPC-UNI reflects a conditional bias in the gridded reference rather than over-correction; independent station validation (Section 3.3) confirms ratios near 1.0 against gauges. Event detection shows a modest trade-off: POD decreases from 0.767 to 0.707 as dry-day frequency matching reclassifies some satellite wet days, while FAR improves from 0.271 to 0.248. Temporal accuracy metrics (Pearson correlation, RMSE, NSE) remain essentially unchanged, consistent with the theoretical constraint that quantile mapping adjusts marginal distributions without modifying temporal sequencing [34].

Table 1. Domain-wide evaluation by correction stage against CPC-UNI. Values represent the spatial median across all land pixels. Ratio metrics express test/reference (target = 1.0). Bold indicates the value closest to the target.

Evaluation Objective	Metric	LS	LSEQM	LSEQM+DL	Target
Value Adjustment	Relative Bias	0.0000	0.0697	0.0652	0

Evaluation Objective	Metric	LS	LSEQM	LSEQM+DL	Target
Distribution Alignment	Std. Dev. Ratio	0.9707	1.1587	1.1475	1.0
	KS p -value (%)	6.56	0.00	0.00	100
	Wet-Day Freq. Ratio	1.0397	0.9466	0.9462	1.0
Extreme Preservation	Wet-Day Int. Ratio	0.9611	1.1569	1.1519	1.0
	Q_{95} Ratio	0.9459	1.1711	1.1626	1.0
	Q_{99} Ratio	0.9669	1.2117	1.1964	1.0
Event Detection	POD	0.7669	0.7069	0.7069	1.0
	CSI	0.5890	0.5723	0.5724	1.0
	FAR	0.2705	0.2476	0.2475	0
Temporal Skill (acknowledged limitation)	Pearson r	0.3430	0.3451	0.3476	1.0
	RMSE (mm/day)	13.105	14.180	14.072	0
	NSE	-0.273	-0.548	-0.524	1.0

3.2 Taylor Diagram Summary

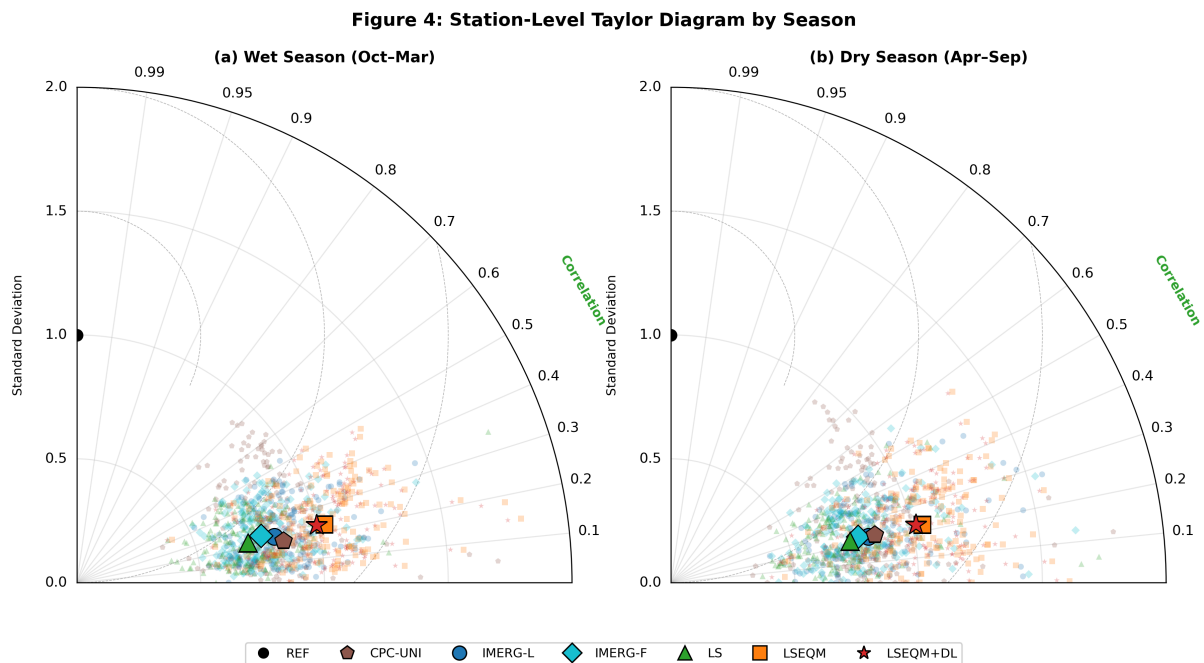


Figure 2: Taylor diagrams comparing daily precipitation performance against independent BMKG station observations, stratified by (a) wet season (October–March) and (b) dry season (April–September). Products: CPC-UNI (brown pentagon), IMERG-L (blue circle), IMERG-F (cyan diamond), LS (green triangle), LSEQM (orange square), LSEQM+DL (red star). Large markers show the per-station median; small translucent markers show individual stations. Angular position: Pearson correlation; radial distance: normalized standard deviation; gray arcs: centered RMSE [35].

Figure 2 provides a compact multi-dimensional summary of correction performance against independent BMKG stations, stratified by monsoon season. In both seasons, LS underestimates observed variability (normalized standard deviation 0.71–0.74), while LSEQM and LSEQM+DL bring the ratio toward unity (1.03/1.00 in the wet season, 1.04/1.02 in the dry season). Pearson correlation remains largely unchanged across correction stages (≈ 0.22 – 0.25), confirming that

distributional correction does not modify temporal prediction skill, consistent with the theoretical expectation in Section 3.1. CPC-UNI itself does not coincide with the station reference point (standard deviation ratio ≈ 0.85), reflecting interpolation uncertainty at 0.5° resolution, a characteristic that motivates the independent station validation in Section 3.3.

3.3 Independent Station Validation

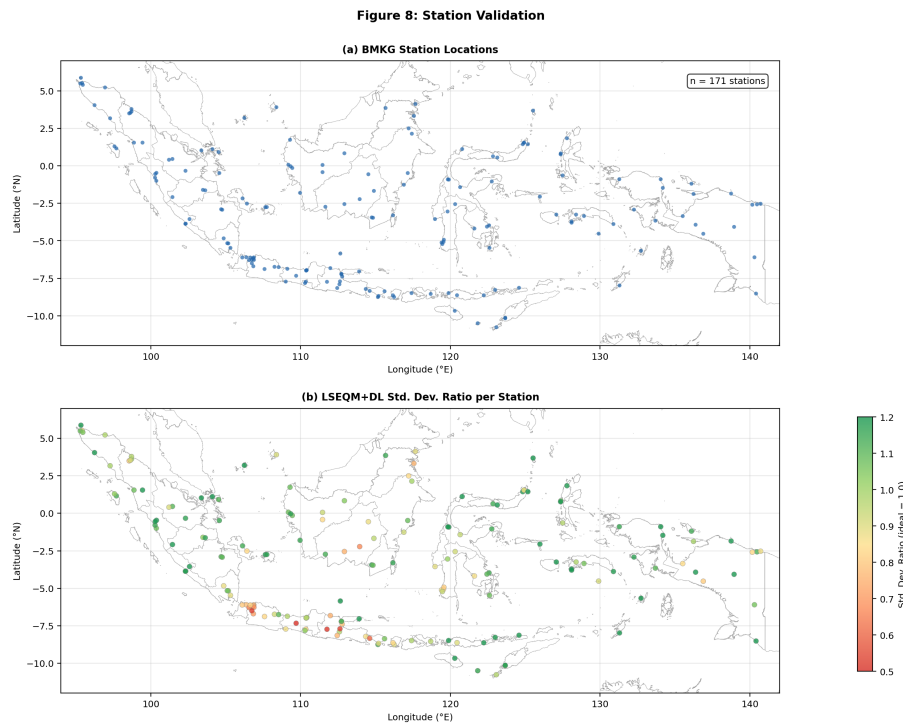


Figure 3: (a) Location of the 171 BMKG stations used for independent validation. (b) Per-station standard deviation ratio (LSEQM+DL vs. observed), averaged across all dekadal periods. Values near 1.0 (green) indicate good variability preservation.

Table 2 presents the median validation results across 171 BMKG stations. These observations are entirely independent of the correction procedure and represent direct ground truth uncontaminated by CPC-UNI interpolation uncertainty.

The LS-to-LSEQM transition delivers the largest improvements. The standard deviation ratio moves from 0.71 to 1.03, correcting a 29% underestimation of precipitation variability to within 3% of observed. LSEQM+DL reaches 1.00 (exact match). The wet-day frequency ratio moves from 1.21 to 0.96, reducing a 21% overestimation of wet-day occurrence to within 5% of observed. The Q_{99} ratio improves from 0.71 to 1.05 (LSEQM) and 1.01 (LSEQM+DL), recovering the large extreme rainfall underestimation present in the LS product. The station-level relative bias improves from -11.4% to -0.6% . These results confirm that the apparent CPC-UNI overshoot in the domain-wide percentile ratios (Section 3.1) represents reference-side interpolation uncertainty, not over-correction.

Table 2. Independent station validation organized by the three design objectives. Values represent the median across 171 BMKG stations and all 36 dekadal periods. Ratio metrics express

172 test/observed (target = 1.0). Bold indicates the value closest to the target.

Evaluation Objective	Metric	LS	LSEQM	LSEQM+DL	Target
Value Adjustment	Relative Bias	−0.114	0.009	−0.006	0
	Std. Dev. Ratio	0.71	1.03	1.00	1.0
Distribution Alignment	KS p -value (%)	0.01	19.07	19.07	100
	Wet-Day Freq. Ratio	1.21	0.96	0.95	1.0
	Wet-Day Int. Ratio	0.71	1.06	1.05	1.0
Extreme Preservation	Q ₉₅ Ratio	0.74	1.07	1.05	1.0
	Q ₉₉ Ratio	0.71	1.05	1.01	1.0
Event Detection	POD	0.78	0.65	0.65	1.0
	CSI	0.53	0.49	0.49	1.0
	FAR	0.36	0.32	0.32	0

173 3.4 Extreme Event Detection

Figure 9: Multi-Threshold Verification (WMO framework)

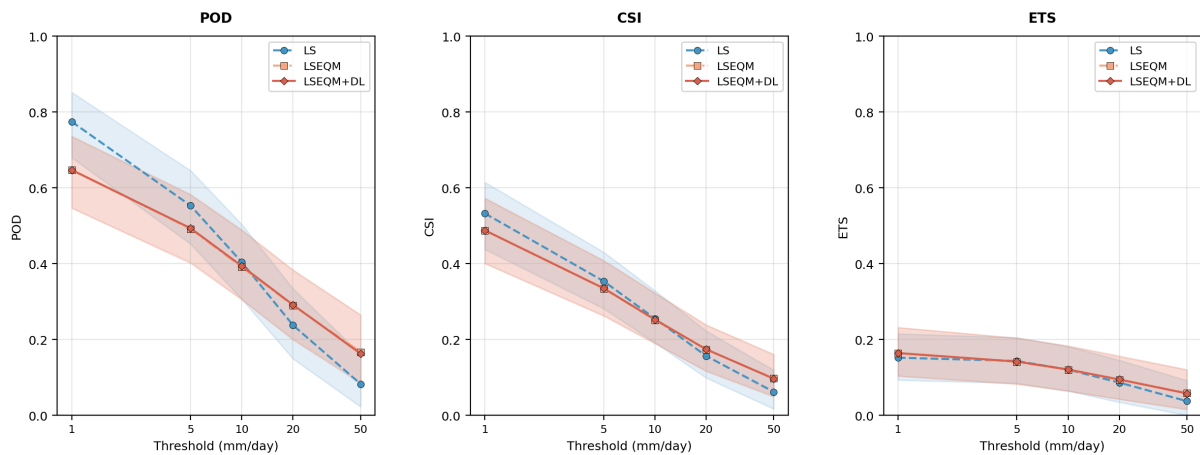


Figure 4: WMO multi-threshold verification at BMKG station locations: (a) Probability of Detection (POD), (b) Critical Success Index (CSI), and (c) Equitable Threat Score (ETS) as functions of precipitation threshold for LS (blue), LSEQM (orange), and LSEQM+DL (green). Lines show the station median; shading shows the interquartile range. Thresholds: 1 mm (measurable), 5 mm (light), 10 mm (moderate), 20 mm (moderate-heavy), 50 mm (heavy), 100 mm (very heavy), 150 mm (extreme).

174 At the 1 mm threshold, LS achieves the highest POD (0.77) and CSI (0.53), as the uniform
175 multiplicative rescaling preserves the satellite wet-day frequency. The pattern reverses above
176 20 mm/day: LSEQM and LSEQM+DL maintain higher detection skill than LS at all higher
177 thresholds. At 50 mm, the median CSI for LSEQM+DL is 0.097 versus 0.062 for LS, a 57%
178 improvement. At 100 mm (very heavy rainfall), LSEQM+DL achieves a median POD of 0.065
179 and CSI of 0.042, compared with 0.015 and 0.011 for LS. The Equitable Threat Score, which
180 accounts for random hits, shows the same pattern: at 50 mm, ETS for LSEQM+DL (0.058)
181 exceeds that of LS (0.038), confirming that the improved extreme detection is not attributable
182 to increased frequency bias.

Figure 11: Monthly Temporal Stability of Corrections

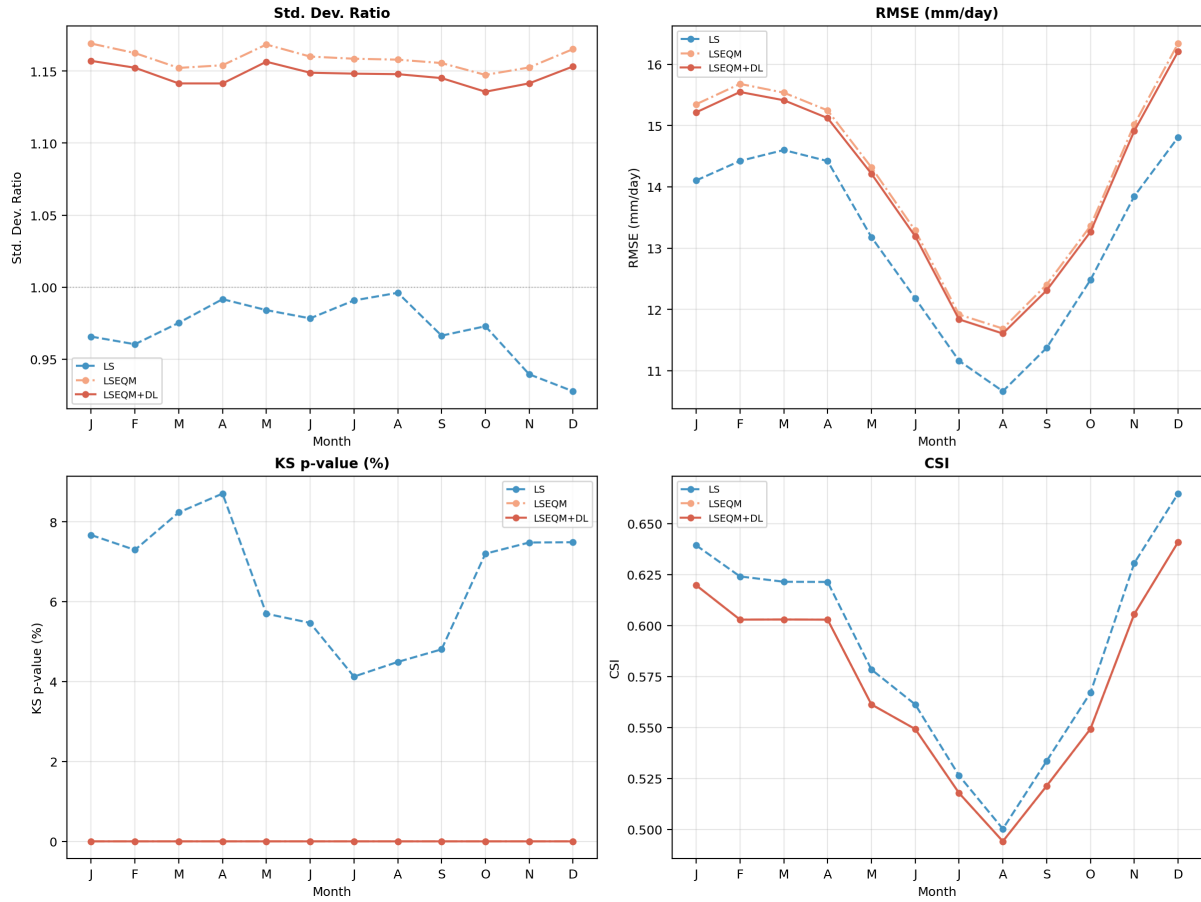


Figure 5: Monthly temporal stability of correction performance: (a) standard deviation ratio, (b) RMSE (mm/day), (c) KS p -value (%), and (d) Critical Success Index (CSI). Lines show domain-median values for LS (blue), LSEQM (orange), and LSEQM+DL (green). The seasonal cycle reflects the wet (October–March) and dry (April–September) monsoon regime.

Figure 5 shows that the method ordering is stable across all 12 calendar months and both monsoon regimes, with no degradation during the wet season when precipitation intensities are highest. The standard deviation ratio remains near 1.0 for LSEQM+DL throughout the year, while LS consistently underestimates variability. RMSE is higher for LSEQM and LSEQM+DL than LS, reflecting the wider corrected distribution rather than degraded temporal accuracy.

4 Discussion and Conclusions

The LSEQM+DL framework delivers improvements across all three design objectives, confirmed by independent validation against BMKG station observations absent from the correction procedure. The staged design assigns each component a specific role: LS removes the mean bias, EQM with gamma fitting aligns the distribution body, GPD explicitly extends the correction into the extreme tail, and CNN refinement addresses spatially structured residuals in

high-intensity pixels. The station density confidence mask ensures that the CNN contribution is proportional to the reliability of the training target.

Three findings warrant emphasis. First, the apparent overshoot of upper percentile ratios against CPC-UNI resolves to near-unity at independent stations (domain Q_{99} ratio 1.196 vs. station Q_{99} ratio 1.01), supporting interpretation of the domain-wide residual as reference-side interpolation uncertainty rather than over-correction. Second, the WMO multi-threshold verification demonstrates that GPD tail adjustment provides measurable skill above 20 mm/day that would be absent from a pure EQM approach. Third, the trade-off between wet-day frequency correction and event detection is explicit, predictable, and a direct consequence of the dry-day frequency matching design [11], not an uncontrolled artifact.

Several limitations apply. The blending weight ($\alpha = 0.70$) and GPD threshold (80th percentile) were selected on domain expertise and not formally optimized; formal sensitivity analysis is a priority for future work. The CNN architecture is simple by design, given the limited training sample (~ 250 values per dekad), and alternatives such as U-Net may improve spatial refinement. No strict temporal hold-out is applied, since per-dekad training pools all available years.

The framework relies on globally available datasets (IMERG, CPC-UNI) and employs transferable statistical methods, supporting application to other tropical and subtropical regions with sparse gauge networks. Each correction stage targets a distinct, verifiable bias dimension, making the framework modular: individual stages can be replaced or recalibrated without redesigning the full pipeline. Ongoing work targets additional Maritime Continent domains and formal evaluation of the confidence mask sensitivity to the saturation count parameter.

References

- [1] Hou, A.Y., Kakar, R.K., Neeck, S., Azarbarzin, A.A., Kummerow, C.D., Kojima, M., Oki, R., Nakamura, K. and Iguchi, T. (2014). The Global Precipitation Measurement Mission. *Bull. Amer. Meteor. Soc.*, 95(5), 701-722. DOI: 10.1175/BAMS-D-13-00164.1
- [2] Huffman, G.J., Bolvin, D.T., Braithwaite, D., Hsu, K.-L., Joyce, R.J., Kidd, C., Nelkin, E.J., Sorooshian, S., Stocker, E.F., Tan, J., Wolff, D.B. and Xie, P. (2020). Integrated Multi-satellite Retrievals for the Global Precipitation Measurement (GPM) Mission (IMERG). *Satellite Precipitation Measurement*, Adv. Global Change Res., 67, 343-353. DOI: 10.1007/978-3-030-24568-9_19
- [3] Maggioni, V., Meyers, P.C. and Robinson, M.D. (2016). A Review of Merged High-Resolution Satellite Precipitation Product Accuracy during the Tropical Rainfall Measuring Mission (TRMM) Era. *J. Hydrometeorol.*, 17(4), 1101-1117. DOI: 10.1175/JHM-D-15-0190.1
- [4] Teutschbein, C. and Seibert, J. (2012). Bias correction of regional climate model simulations for hydrological climate-change impact studies: Review and evaluation of different methods. *J. Hydrol.*, 456-457, 12-29. DOI: 10.1016/j.jhydrol.2012.05.052
- [5] Gudmundsson, L., Bremnes, J.B., Haugen, J.E. and Engen-Skaugen, T. (2012). Technical Note: Downscaling RCM precipitation to the station scale using statistical transformations – a comparison of methods. *Hydrol. Earth Syst. Sci.*, 16, 3383-3390. DOI: 10.5194/hess-16-3383-2012
- [6] Themessl, M.J., Gobiet, A. and Heinrich, G. (2012). Empirical-statistical downscaling and error correction of regional climate models and its impact on the climate change signal. *Climatic Change*, 112, 449-468. DOI: 10.1007/s10584-011-0224-4

- [7] Coles, S. (2001). *An Introduction to Statistical Modeling of Extreme Values*. Springer Series in Statistics. DOI: 10.1007/978-1-4471-3675-0
- [8] Maraun, D. (2016). Bias Correcting Climate Change Simulations – a Critical Review. *Curr. Clim. Change Rep.*, 2, 211-220. DOI: 10.1007/s40641-016-0050-x
- [9] Bano-Medina, J., Manzanar, R. and Gutierrez, J.M. (2020). Configuration and intercomparison of deep learning neural models for statistical downscaling. *Geosci. Model Dev.*, 13, 2109-2124. DOI: 10.5194/gmd-13-2109-2020
- [10] Pan, B., Hsu, K., AghaKouchak, A. and Sorooshian, S. (2019). Improving Precipitation Estimation Using Convolutional Neural Network. *Water Resour. Res.*, 55(3), 2301-2321. DOI: 10.1029/2018WR024090
- [11] Cannon, A.J., Sobie, S.R. and Murdock, T.Q. (2015). Bias Correction of GCM Precipitation by Quantile Mapping: How Well Do Methods Preserve Changes in Quantiles and Extremes? *J. Climate*, 28(17), 6938-6959. DOI: 10.1175/JCLI-D-14-00754.1
- [12] Ehret, U., Zehe, E., Wulfmeyer, V., Warrach-Sagi, K. and Liebert, J. (2012). HESS Opinions: Should we apply bias correction to global and regional climate model data? *Hydrol. Earth Syst. Sci.*, 16, 3391-3404. DOI: 10.5194/hess-16-3391-2012
- [13] Chen, M., Shi, W., Xie, P., Silva, V.B.S., Kousky, V.E., Wayne Higgins, R. and Janowiak, J.E. (2008). Assessing objective techniques for gauge-based analyses of global daily precipitation. *J. Geophys. Res.*, 113, D04110. DOI: 10.1029/2007JD009132
- [14] Ebert, E.E. (2007). Methods for Verifying Satellite Precipitation Estimates. *Measuring Precipitation from Space*, Adv. Global Change Res., 28, 345-356. DOI: 10.1007/978-1-4020-5835-6_27
- [15] Wilks, D.S. (2011). *Statistical Methods in the Atmospheric Sciences*. 3rd ed. Academic Press, International Geophysics Series, Vol. 100. DOI: 10.1016/B978-0-12-385022-5.00001-4
- [16] Piani, C., Haerter, J.O. and Coppola, E. (2010). Statistical bias correction for daily precipitation in regional climate models over Europe. *Theor. Appl. Climatol.*, 99, 187-192. DOI: 10.1007/s00704-009-0134-9
- [17] LeCun, Y., Bengio, Y. and Hinton, G. (2015). Deep learning. *Nature*, 521, 436-444. DOI: 10.1038/nature14539
- [18] Xie, P., Chen, M., Yang, S., Yatagai, A., Hayasaka, T., Fukushima, Y. and Liu, C. (2007). A Gauge-Based Analysis of Daily Precipitation over East Asia. *J. Hydrometeor.*, 8(3), 607-626. DOI: 10.1175/JHM583.1
- [19] Aldrian, E. and Susanto, R.D. (2003). Identification of three dominant rainfall regions within Indonesia and their relationship to sea surface temperature. *Int. J. Climatol.*, 23(12), 1435-1452. DOI: 10.1002/joc.950
- [20] As-syakur, A.R., Tanaka, T., Osawa, T. and Mahendra, M.S. (2013). Indonesian rainfall variability observation using TRMM multi-satellite data. *Int. J. Remote Sens.*, 34(21), 7723-7738. DOI: 10.1080/01431161.2013.826837
- [21] Pradhan, R.K., Markonis, Y., Varber, M.R., Franch, M., Papalexiou, S.M., Vidal, J.-P., Pappas, C., Hanel, M. and Hao, Z. (2022). Review of GPM IMERG performance: A global perspective. *Remote Sens. Environ.*, 268, 112754. DOI: 10.1016/j.rse.2021.112754
- [22] Sun, Q., Miao, C., Duan, Q., Ashouri, H., Sorooshian, S. and Hsu, K.-L. (2018). A Review of Global Precipitation Data Sets: Data Sources, Estimation, and Intercomparisons. *Rev. Geophys.*, 56(1), 79-107. DOI: 10.1002/2017RG000574
- [23] Wood, A.W., Leung, L.R., Sridhar, V. and Lettenmaier, D.P. (2004). Hydrologic implications of dynamical and statistical approaches to downscaling climate model outputs. *Climatic Change*, 62, 189-216. DOI: 10.1023/B:CLIM.0000013685.99609.9e
- [24] WMO (2008). *Guide to Meteorological Instruments and Methods of Observation*. WMO-No. 8, World Meteorological Organization, Geneva.
- [25] Hosking, J.R.M. and Wallis, J.R. (1997). *Regional Frequency Analysis: An Approach Based on L-Moments*. Cambridge University Press. DOI: 10.1017/CBO9780511529443
- [26] Kingma, D.P. and Ba, J. (2015). Adam: A Method for Stochastic Optimization. *Proc. 3rd Int. Conf. Learning Representations (ICLR 2015)*, San Diego.
- [27] Mega, T., Ushio, T., Matsuda, T., Kubota, T., Kachi, M. and Oki, R. (2019). Gauge-Adjusted Global Satellite Mapping of Precipitation. *IEEE Trans. Geosci. Remote Sens.*, 57(4), 1928-1935. DOI: 10.1109/TGRS.2018.2870199

- [28] Nash, J.E. and Sutcliffe, J.V. (1970). River flow forecasting through conceptual models part I – A discussion of principles. *J. Hydrol.*, 10(3), 282-290. DOI: 10.1016/0022-1694(70)90255-6
- [29] Moriasi, D.N., Arnold, J.G., Van Liew, M.W., Bingner, R.L., Harmel, R.D. and Veith, T.L. (2007). Model evaluation guidelines for systematic quantification of accuracy in watershed simulations. *Trans. ASABE*, 50(3), 885-900. DOI: 10.13031/2013.23153
- [30] Gupta, H.V., Kling, H., Yilmaz, K.K. and Martinez, G.F. (2009). Decomposition of the mean squared error and NSE performance criteria: Implications for improving hydrological modelling. *J. Hydrol.*, 377(1-2), 80-91. DOI: 10.1016/j.jhydrol.2009.08.003
- [31] Tan, J., Petersen, W.A. and Tokay, A. (2016). A Novel Approach to Identify Sources of Errors in IMERG for GPM Ground Validation. *J. Hydrometeor.*, 17(9), 2477-2491. DOI: 10.1175/JHM-D-16-0079.1
- [32] Ramadhan, R., Yusnaini, H., Marzuki, M., Muharsyah, R., Suryanto, W., Sholihun, S., Vonnisa, M., Harmadi, H., Ningsih, A.P., Battaglia, A., Hashiguchi, H. and Tokay, A. (2022). Evaluation of GPM IMERG Performance Using Gauge Data over Indonesian Maritime Continent at Different Time Scales. *Remote Sensing*, 14(5), 1172. DOI: 10.3390/rs14051172
- [33] Tang, G., Clark, M.P., Papalexiou, S.M., Ma, Z. and Hong, Y. (2020). Have satellite precipitation products improved over last two decades? A comprehensive comparison of GPM IMERG with nine satellite and reanalysis datasets. *Remote Sens. Environ.*, 240, 111697. DOI: 10.1016/j.rse.2020.111697
- [34] Maraun, D. (2013). Bias correction, quantile mapping, and downscaling: Revisiting the inflation issue. *J. Climate*, 26(6), 2137-2143. DOI: 10.1175/JCLI-D-12-00821.1
- [35] Taylor, K.E. (2001). Summarizing multiple aspects of model performance in a single diagram. *J. Geophys. Res.*, 106(D7), 7183-7192. DOI: 10.1029/2000JD900719
- [36] Adam, J.C. and Lettenmaier, D.P. (2003). Adjustment of global gridded precipitation for systematic bias. *J. Geophys. Res.*, 108(D9), 4257. DOI: 10.1029/2002JD002499
- [37] Behrangi, A., Khakbaz, B., Jaw, T.C., AghaKouchak, A., Hsu, K. and Sorooshian, S. (2011). Hydrologic evaluation of satellite precipitation products over a mid-size basin. *J. Hydrol.*, 397(3-4), 225-237. DOI: 10.1016/j.jhydrol.2010.11.043
- [38] Rahmawati, N. and Lubczynski, M.W. (2018). Validation of satellite daily rainfall estimates in complex terrain of Bali Island, Indonesia. *Theor. Appl. Climatol.*, 134(1-2), 513-532. DOI: 10.1007/s00704-017-2290-7
- [39] Wang, Z., Zhong, R., Lai, C. and Chen, J. (2017). Evaluation of the GPM IMERG satellite-based precipitation products and the hydrological utility. *Atmos. Res.*, 196, 151-163. DOI: 10.1016/j.atmosres.2017.06.020
- [40] Sha, Y., Gagne II, D.J., West, G. and Stull, R. (2020). Deep-learning-based gridded downscaling of surface meteorological variables in complex terrain. Part I: Daily maximum and minimum 2-m temperature. *J. Appl. Meteor. Climatol.*, 59(12), 2057-2073. DOI: 10.1175/JAMC-D-20-0057.1
- [41] Vandal, T., Kodra, E., Ganguly, S., Michaelis, A., Nemani, R. and Ganguly, A.R. (2017). DeepSD: Generating high fidelity daily climate surfaces with generative adversarial networks. *Proc. 23rd ACM SIGKDD Int. Conf. on Knowledge Discovery and Data Mining*, 1375-1383. DOI: 10.1145/3097983.3098004
- [42] Beck, H.E., Wood, E.F., Pan, M., Fisher, C.K., Miralles, D.G., van Dijk, A.I.J.M., McVicar, T.R. and Adler, R.F. (2019). MSWEP V2 Global 3-Hourly 0.1 degrees Precipitation: Methodology and Quantitative Assessment. *Bull. Amer. Meteor. Soc.*, 100(3), 473-500. DOI: 10.1175/BAMS-D-17-0138.1
- [43] Lange, S. (2019). Trend-preserving bias adjustment and statistical downscaling with ISIMIP3BASD (v1.0). *Geosci. Model Dev.*, 12, 3055-3070. DOI: 10.5194/gmd-12-3055-2019
- [44] Hempel, S., Frieler, K., Warszawski, L., Schewe, J. and Piontek, F. (2013). A trend-preserving bias correction – the ISI-MIP approach. *Earth Syst. Dynam.*, 4, 219-236. DOI: 10.5194/esd-4-219-2013
- [45] Baez-Villanueva, O.M., Zambrano-Bigiarini, M., Beck, H.E., McNamara, I., Ribbe, L., Nauditt, A., Birkel, C., Verbist, K., Giraldo-Osorio, J.D. and Thinh, N.X. (2020). RF-MEP: A novel Random Forest method for merging gridded precipitation products and ground-based measurements. *Remote Sens. Environ.*, 239, 111606. DOI: 10.1016/j.rse.2019.111606

- 330 [46] Sillmann, J., Kharin, V.V., Zhang, X., Zwiers, F.W. and Bronaugh, D. (2013). Climate extremes indices in the
331 CMIP5 multimodel ensemble: Part 1. Model evaluation in the present climate. *J. Geophys. Res. Atmos.*, 118,
332 1716-1733. DOI: 10.1002/jgrd.50203
- 333 [47] Wu, P., Yamanaka, M.D. and Matsumoto, J. (2008). The Formation of Nocturnal Rainfall Offshore from Convec-
334 tion over Western Kalimantan (Borneo) Island. *J. Meteor. Soc. Japan*, 86A, 187-203. DOI: 10.2151/jmsj.86A.187